

Interacting Multipoles of Rank 1 and 1:

$$T_{\alpha\beta}\mu_{\alpha}^A\mu_{\beta}^B = R_z^{-5} \cdot \begin{bmatrix} +3 \cdot R_{\alpha} \cdot R_{\beta} \cdot \mu_{\alpha}^A \cdot \mu_{\beta}^B \\ -1 \cdot R_z^2 \cdot \delta(\alpha, \beta) \cdot \mu_{\alpha}^A \cdot \mu_{\beta}^B \end{bmatrix} \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta}\mu_{\alpha}^A\mu_{\beta}^B = R_z^{-5} \cdot \begin{bmatrix} +3 \cdot R_{\alpha} \cdot R_{\beta} \cdot \mu_{\alpha}^A \cdot \mu_{\beta}^B \\ -1 \cdot R_z^2 \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \mu_{\alpha}^B \end{bmatrix} \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot \begin{bmatrix} +3 \cdot R_{\alpha} \cdot R_{\beta} \cdot \mu_{\alpha}^A \cdot \mu_{\beta}^B \\ -1 \cdot R_z^2 \cdot \mu_{\alpha}^A \cdot \mu_{\alpha}^B \end{bmatrix} \quad (2b)$$

Simplify R:

$$T_{\alpha\beta}\mu_{\alpha}^A\mu_{\beta}^B = R_z^{-5} \cdot \begin{bmatrix} +3 \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \mu_z^B \\ -1 \cdot R_z^2 \cdot \mu_{\alpha}^A \cdot \mu_{\alpha}^B \end{bmatrix} \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_{\alpha}^A \cdot \mu_{\alpha}^B \end{bmatrix} \quad (3b)$$

Simplify Dipoles:

$$T_{\alpha\beta}\mu_{\alpha}^A\mu_{\beta}^B = R_z^{-5} \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y^A \cdot \mu_y^B \end{bmatrix} \quad (4)$$

Combining everything:

$$T_{\alpha\beta}\mu_{\alpha}^A\mu_{\beta}^B = \begin{bmatrix} -1 \cdot R_z^{-3} \cdot \mu_y^A \cdot \mu_y^B \end{bmatrix} \quad (5)$$